

LAB 6—FINDING THE INDUCTANCE OF A HELMHOLTZ COIL

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1 Abstract

Helmholtz coils are used in lab settings to induce a uniform B-field in a region between two coils of wire when a current is run through them. These coils are set up in such a way that the Helmholtz coil has a measurable self-inductance when set up in a circuit with a changing voltage and other components. In this lab, we measured the current of a system with a resistor and a direct voltage supply, and we used this current to determine the self-inductance of the Helmholtz coil. Our model of the current of the system as the voltage intermittently turned on and off aligned well with the experimental data, and our calculation of the inductance will be useful for future experiments using similar Helmholtz coils.

2 Background

Starting from the Kirchoff loop rule, the charging condition, where the voltage from the voltage source is $\mathcal{E}$, is given by:

$$\mathcal{E} - IR - L \frac{\partial I}{\partial t} = 0$$

Where I represents the current, $\mathcal{E}$ represents the electromotive force, R represents the total resistance in the circuit, L represents the inductance, and t represents the time. Solving this equation for I , we find that the current in the circuit after the voltage is turned on, in the “charging” stage, can be calculated from the following:

$$I(t) = \frac{\mathcal{E}}{R} (1 - e^{-\frac{R}{L}t}) \tag{1}$$

Again starting from the Kirchoff loop rule, the discharging condition, where the voltage from the voltage source is 0, is given by:

$$-IR - L \frac{\partial I}{\partial t} = 0$$

Solving this equation for I , we find that the current in the circuit after the voltage is turned off, in the “discharging” stage, can be calculated from the following:

$$I(t) = \frac{\mathcal{E}}{R} e^{-\frac{R}{L}t} \quad (2)$$

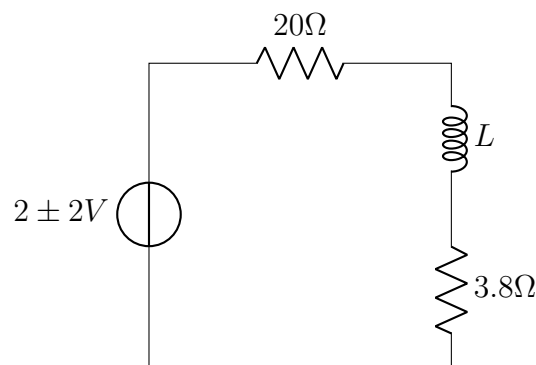


Figure 1: A basic diagram of the circuit. The two components on the right side represent both the internal resistance and the inductance of the Helmholtz coil

3 Materials

1. A table
2. A Helmholtz coil
3. A multimeter
4. A function generator
5. A variable resistor
6. A current probe
7. Assorted wires and alligator clips
8. A LabQuest connected to a computer with LoggerPro

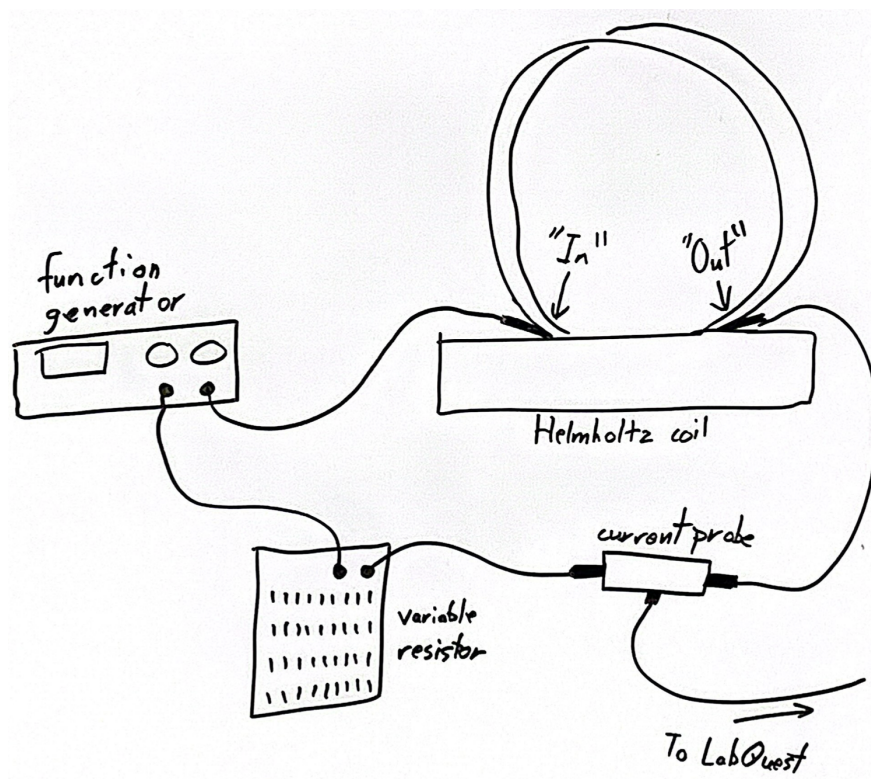


Figure 2: The experimental setup

4 Procedure

4.1 Setup

1. We attached the multimeter in resistance measuring mode to the variable resistor and set it to as close to $20\ \Omega$ as possible using its panel of switches.
2. We measured the resistance through the Helmholtz coil also using the multimeter.
3. We attached the black lead of the function generator to the “In” of the front loop of the Helmholtz coil using an alligator clip and ran a wire from the “Out” of the Helmholtz coil to the current probe, from the current probe to the variable resistor, and from the variable resistor back to the red lead of the function generator. (See the full setup diagram in Figure 2)

4.2 Experiment

1. We set the function generator to a square wave with an amplitude of 2 volts, an offset of +2 volts, and a frequency of 8 Hz.

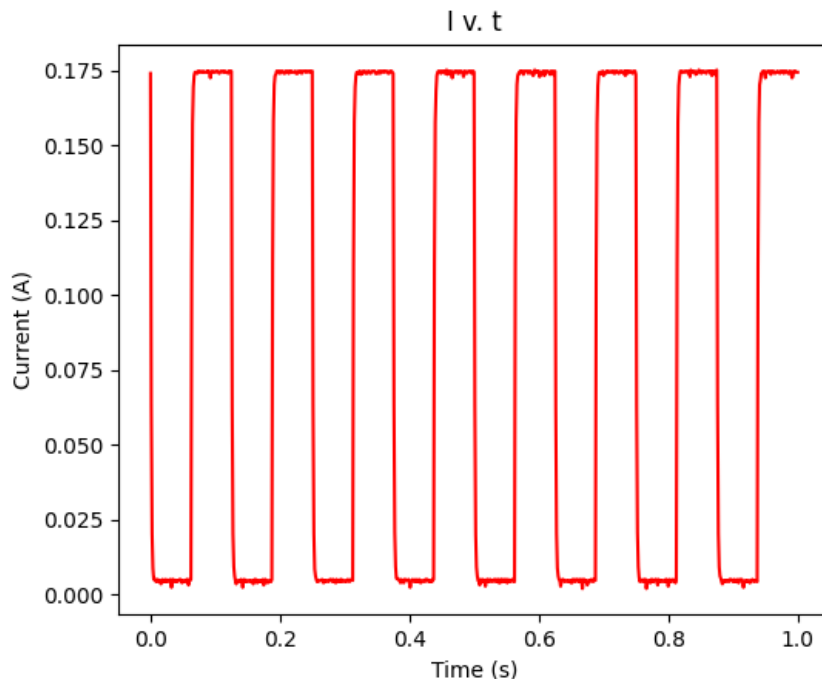


Figure 3: The experimental data

2. We set the sampling rate for the current probe in LoggerPro to 800 samples/second with a duration of 1 second.
3. We recorded the current across the entire circuit while the voltage was switching between 0 and 4 volts. The results are below in Figure 3.

5 Data and Analysis

By using a square wave, we turned the circuit off and on 8 times a second. The resistance across the two components was measured as $R_{\text{coil}} = 3.8 \pm 0.1 \, \Omega$ and $R_{\text{resistor}} = 20.0 \pm 0.5 \, \Omega$. We took the data and separated them into their distinct charging and discharging sections. We then fit our data to their respective equations from the Purpose section above:

$$\text{Charging: } I(t) = \frac{\mathcal{E}}{R}(1 - e^{-\frac{R}{L}t}) \qquad \text{Discharging: } I(t) = \frac{\mathcal{E}}{R}e^{-\frac{R}{L}t}$$

By our fits, we got the results that can be seen in Figure 4 and Table 1.

As we can see from the fit in Figure 3 and their respective R^2 values in Table 1, the fit for the charging portion of the curve is more accurate than the discharging portion of the curve. This might be because, as we can see in the data and from the voltage found for the fit, the square wave that the function generator generated might have oscillated between values closer to 0.15 to 4.15 volts, despite being set to an amplitude of 2 volts and an offset of 2 volts, which should have resulted in a voltage that turned on to 4 volts and off to 0 volts.

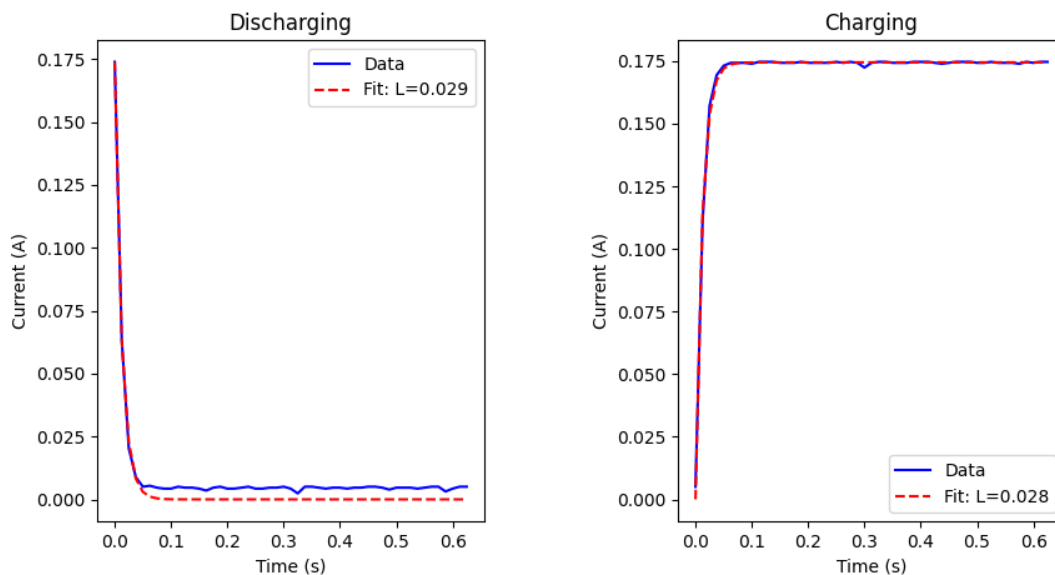


Figure 4: One fit of current with respect to time

L_D	R_D^2	V_D	L_C	R_C^2	V_C
0.0292 ± 0.0004	0.969	4.14 ± 0.1	0.0280 ± 0.0004	0.998	4.15 ± 0.004
0.0290 ± 0.0004	0.970	4.15 ± 0.1	0.0279 ± 0.0004	0.999	4.15 ± 0.004
0.0292 ± 0.0004	0.969	4.16 ± 0.1	0.0281 ± 0.0004	0.998	4.15 ± 0.004
0.0294 ± 0.0004	0.969	4.14 ± 0.1	0.0281 ± 0.0004	0.998	4.15 ± 0.004
0.0291 ± 0.0004	0.969	4.15 ± 0.1	0.0279 ± 0.0004	0.998	4.15 ± 0.004
0.0289 ± 0.0004	0.970	4.15 ± 0.1	0.0283 ± 0.0004	0.998	4.15 ± 0.004
0.0293 ± 0.0004	0.970	4.13 ± 0.1	0.0280 ± 0.0004	0.998	4.15 ± 0.004
0.0293 ± 0.0004	0.970	4.16 ± 0.1	0.0281 ± 0.0004	0.998	4.15 ± 0.004

Table 1: Data from fitting with `scipy.optimize.curve_fit`, wherein L_D is the inductance, V_D is the voltage, and R_D^2 is the R^2 from the fit of the discharging data, and wherein L_C is the inductance, V_C is the voltage, and R_C^2 is the R^2 from the fit of the charging data

Then, since the resistance measured by instrumental error was $R_{total} = 23.8 \pm 0.6 \Omega$, to find the error in the result, the data was fitted to the equation above using $R_{total} = 23.8 - 0.6 = 23.2 \Omega$ and $R_{total} = 23.8 + 0.6 = 24.4 \Omega$, giving a final $L = 0.028 \pm 0.006$ Henries after accounting for this difference and for the standard deviation between the measurements.

6 Conclusion

In conclusion, the inductance of the coil, measured by this experiment, using the charging fits (and propagating the error), is $L = 0.028 \pm 0.006$ Henries. The expected value of the coil, using its geometry, is about $L = 0.020$ Henries, so there is about a 40% difference

mostly due to precision errors. In order to improve this offset, we would need to measure the current and resistance more accurately. The resistance through the variable resistor was more variable than intended, as the reading from the multimeter changed when disconnected and reconnected when none of the switches were flipped.